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0.1 Exersice

Exercise 0.1.1 ▸ Calculate

Let V be a vector space of finite dimension with basis $(e_1, \dots, e_n)$ and let V^* denotes the dual space with dual basis $(e_1^*, \dots, e_n^*)$. Please prove in two different ways that for every $(\alpha, \beta) \in \Lambda^p V^* \times \Lambda^q V^*$,

$$\alpha \wedge \beta = (-1)^{pq} \beta \wedge \alpha.$$

One way is to use definition, the other is to calculate in basis using the next exercise.

Proof. 1.

$$\alpha \wedge \beta = \frac{(p+q)!}{p!q!} \mathrm{Alt}\,(\alpha \otimes \beta)$$

$$\beta \wedge \alpha = \frac{(p+q)!}{p!q!} \mathrm{Alt}\,(\beta \otimes \alpha)$$

Only need to show that

$$\mathrm{Alt}\,(\alpha \otimes \beta) = (-1)^{pq}\,(\beta \otimes \alpha)$$

Where

$$\mathrm{Alt}\,(\alpha \otimes \beta) = \frac{1}{(p+q)!} \sum_{\sigma \in S_{p+q}} (\mathrm{sgn}\sigma)\,\sigma\,(\alpha \otimes \beta)$$

Let $\tau = (p+1, \cdots, p+q, 1, 2, \cdots, p)$, then $\operatorname{sgn}(\tau) = (-1)^{pq}$. We have

$$\begin{aligned}
\operatorname{Alt}(\alpha \otimes \beta) &= \frac{1}{(p+q)!} \sum_{\sigma \in S_{p+q}} (\operatorname{sgn}\sigma)\, \sigma \circ \tau (\beta \otimes \alpha) \\
&= (-1)^{pq} \frac{1}{(p+q)!} \sum_{\sigma \in S_{p+q}} (\operatorname{sgn}\sigma)(\operatorname{sgn}\tau)(\sigma \circ \tau)(\beta \otimes \alpha) \\
&= (-1)^{pq} \frac{1}{(p+q)!} \sum_{\sigma' \in S_{p+q}} (\operatorname{sgn}\sigma')\, \sigma' (\beta \otimes \alpha) \\
&= (-1)^{pq} \operatorname{Alt}(\beta \otimes \alpha)
\end{aligned}$$

2. Suppose

$$\alpha = \sum_{i_1 \cdots i_p} a_{i_1 \cdots i_p} e^*_{i_1} \wedge \cdots \wedge e^*_{i_p}$$

$$\beta = \sum_{j_1 \cdots j_q} b_{j_1 \cdots j_q} e^*_{j_1} \wedge \cdots \wedge e^*_{j_q}$$

Then

$$\alpha \wedge \beta = \sum_{i_1 \cdots i_p} a_{i_1 \cdots i_p} b_{j_1 \cdots j_q} (e^*)_I \wedge (e^*)_J = \sum_{I,J} a_I b_J e_{IJ}$$

$$\beta \wedge \alpha = \sum_{I,J} a_I b_J e_{JI} = (-1)^{pq} \sum_{I,J} a_I b_J e_{IJ} = (-1)^{pq} \alpha \wedge \beta$$

□

Exercise 0.1.2

Let V be a vector space of finite dimension and let V^* denotes the dual space. Prove that if $(\theta_1, \ldots, \theta_r)$ is a family of r covectors in V^* which are ***linearly dependent***, then the following wedge product:

$$\theta_1 \wedge \cdots \wedge \theta_r = 0.$$

Proof. Since $\theta_1 \wedge \cdots \theta_r$ is multi-linear and alternative, take any vector $v^1, \cdots, v^r$, we have

$$\theta_1 \wedge \cdots \wedge \theta_r \left(v^1, \cdots, v^r\right) = \det\left(\theta_i\left(v^j\right)\right)_{ij}$$

Since $\theta_1, \cdots, \theta_r$ are linealy dependent , the row vectors of the matrix $\left(\theta_i\left(v^j\right)\right)_{ij}$ are as well. Thus $\det\left(\theta_i\left(v^j\right)\right)_{ij} = 0$, and $\theta_1 \wedge \cdots \theta_r\left(v^1, \cdots, v^r\right) = 0$. Since $v^1, \cdots, v^r$ are arbitrary , $\theta_1 \wedge \cdots \wedge \theta_r = 0$. □

Exercise 0.1.3 ▸ Basis of Grassmann algebra

Let V be a vector space of finite dimension with basis $(e_1, \ldots, e_n)$ and let V^* denotes the dual space with dual basis $(e_1^*, \ldots, e_n^*)$. Please prove that

$$(e_{i_1}^* \wedge \cdots \wedge e_{i_k}^*)_{1 \le i_1 < \cdots < i_k \le n}$$

forms a basis of $\Lambda^k V^*$.

Proof. For $\alpha \in \Lambda^k V^*$, and $I = (i_1, \cdots, i_k)$. Define

$$\alpha_I := \alpha\left(e_{i_1}, \cdots, e_{i_k}\right)$$

If I has a reputed index , then $\alpha_I = 0$, and if $J = \sigma I$, $\alpha_J = (\mathrm{sgn}\sigma)\,\alpha_I$. We have

$$\sum_I{}' \alpha_I e_I^*\left(e_{j_1}, \cdots, e_{j_k}\right) = \sum_I{}' \alpha_I \delta_J^I = \alpha_J = \alpha\left(e_{j_1}, \cdots, e_{j_k}\right)$$

Thus $\alpha = \sum_I' \alpha_I e_I^*$. Which shows that $e_{i_1}^*, \cdots, e_{i_k}^*, 1 \le i_1 < \cdots < i_k \le n$ spands the $\Lambda^k V^*$. To show that they are linealy independent. Supppose that

$$\sum_I k_I e_I^* = 0$$

Both sides act on e_J, we have $k_J = 0$. □

Exercise 0.1.4 ▸ Grassmann algebra versus Polynomial functions on V.

Let V be a vector space of finite dimension and let V^* denotes the dual space. A p-multilinear map α on V is ***symmetric*** if

$$\alpha(v_1, \ldots, v_n) = \alpha(v_{\sigma(1)}, \ldots, v_{\sigma(n)})$$

for all permutations σ. Such α defines a homogeneous polynomial $\tilde{\alpha}$

defined by

$$\tilde{\alpha}(v) = \alpha(v, \dots, v)$$

for all $v \in V$. Then show that for symmetric multilinear maps α, β on V of respective p, q, the product $\alpha \circ \beta$ defined by

$$\alpha\circ\beta(v_1, \dots, v_{p+q}) := \frac{1}{p!q!} \sum_{\sigma\in S_{p+q}} \alpha(v_{\sigma(1)}, \dots, v_{\sigma(p)})\beta(v_{\sigma(p+1)}, \dots, v_{\sigma(p+q)})$$

is valued into symmetric multilinear map on V of degree $p+q$ and that

$$\alpha \circ \beta(v, \dots, v) = \alpha(v, \dots, v)\beta(v, \dots, v).$$

Could you compare with $\Lambda^{\bullet}V^*, \wedge$?

Exercise 0.1.5 ▶ Basic important examples of smooth manifolds

Please prove that the following topological spaces have a smooth manifold structure :

- The Euclidean sphere $S^n \subset \mathbb{R}^{n+1}$.
- The matrix groups $GL_n(\mathbb{R}), SL_n(\mathbb{R}), O_n(\mathbb{R})$, these are Lie groups of matrice. Each time could you tell me what is the dimension of these groups.

Proof.
- Define F:

$$F\left(x^1, \cdots, x^{n+1}\right) = x_1^2 + \cdots + x_{n+1}^2$$

Then

$$S^n = F^{-1}(1)$$

$$\mathrm{d}F_x = \nabla F(x) = \left(2x_1, \cdots, 2_{x_{n+1}}\right) \neq 0, \forall x \in \mathbb{R}^{n+1}$$

By regular value theorem, $S^n \subseteq \mathbb{R}^{n+1}$ is a regular embeded submanifold of $\mathbb{R}^{n+1}$, which itself is a manifold.

- 1. $\mathrm{Mat}(n, \mathbb{R}) \simeq \mathbb{R}^2$ 是 n^2 维流形.
  2. $\mathrm{GL}(n, \mathbb{R}) = \{M \in \mathrm{Mat}(n, \mathbb{R}) : \det M \neq 0\}$ 作为 $\mathrm{Mat}(n, \mathbb{R})$ 的一个开子集也是一个 n^2 维流形.

3. $\mathrm{SL}(n,\mathbb{R}) = (M \in \mathrm{Mat}(n,\mathbb{R}))$ ∶ $\det M = 1$

$$\begin{aligned} \det : \mathrm{Mat}(n,\mathbb{R}) &\to \mathbb{R} \\ M &\mapsto \det M \end{aligned}$$

$SL(n,\mathbb{R}) = \det^{-1}(I)$. 取一个 $M \in \mathrm{Mat}(n,\mathbb{R})$, $\det M = 1$. 要证明 $(\mathrm{d}(\det))_M : \mathbb{R}^{n^2} \to \mathbb{R}$. 按分量求偏导, 得到

$$\frac{\partial}{\partial a_{ij}}(\det M) = A_{ij}(\text{代数余子式})$$

由于 $\det M = 1$. 至少有一个 $A_{ij} \neq 0$. 从而 $(\mathrm{d}(\det))_M$ 是满的.

4. 定义 $P : \mathrm{Mat}(n,\mathbb{R}) \to \mathrm{SMat}(n,\mathbb{R})$(对称矩阵) $\simeq \mathbb{R}^{\frac{n(n+1)}{2}}$.

$$P(A) := A^\top A$$

则 $O(n,\mathbb{R}) = P^{-1}(I)$. 对于 $A \in O(n,\mathbb{R})$.

$$(\mathrm{d}P)_A : \mathrm{Mat}(n,\mathbb{R}) \to \mathrm{SMat}(n,\mathbb{R})$$

与上例不同, 这里用分量的形式去计算 Jacobi 是看不清楚的, 不过由于 P 本身足够简单, 我们可以用微分最原始的含义来观察. 考虑

$$P(A+\varepsilon B) - P(A) = \varepsilon(\mathrm{d}P)_A(B) + O(\cdots)$$

带入得到

$$\left(A^T + \varepsilon B^T\right)(A+\varepsilon B) - A^\top A = \varepsilon\left(B^\top A + A^\top B\right) + O(\varepsilon)$$

于是 $(\mathrm{d}P)_A(B) = B^\top A + A^\top B$. 只需要证明这个关于 B 的映射是满的. 为此, 解方程

$$A^\top B + B^\top A = C$$

取 $B = \frac{1}{2}AC$, 利用 A, C 对称, 带入得到

$$\frac{1}{2}A^\top AC + \frac{1}{2}C^\top A^\top A = \frac{1}{2}\left(C + C^\top\right) = C$$

故 $O(n,\mathbb{R})$ 是 $\mathrm{Mat}(n,\mathbb{R})$ 的一个 $\frac{n(n-1)}{2}$ 维正则子流形. 注意到对于 $A \in O(n,\mathbb{R})$, $\det A = 1$ 或 $\det A = -1$, 可以窥见事实上 $O(n,\mathbb{R})$ 有两个连通分支.

□

Exercise 0.1.6 ▸ Quotient manifolds

Prove that the following objects are quotient manifolds

- $\mathbb{R}P^n$ obtained by quotienting S^n by antipodal map, show that it is nonorientable by a picture. Could you explain by a picture that the neighborhood of the line at infinity is homeomorphic to a Möbius ribbon ?
- Prove that the torus $\mathbb{T}^d = \mathbb{R}^d/\mathbb{Z}^d$ is a smooth manifold.

Proof.

- 1. See $\mathbb{R}P^n$ is $S^2/(x \sim -x)$. The orentation changes at the equator. The map $A : S^2 \to S^2.A(x) = -x$ induces an idendity map on $\mathbb{R}P^2$, $\mathrm{d}A$ send an orentation basis $\{e_1, e_2\}$ to one opposite to it .
  2. Use $\mathbb{R}P^2 = D^2/\{x \sim -x \text{ for } x \in \partial D^2 = S^1\}$. Then the neighborhood of the line at infinity is a thin ring near ∂D^2 with some quotient relation. The quotient is just the same at that of Möbius .

□

Exercise 0.1.7 ▸ Vector fields are derivation

Let M be a C^∞ manifold. Show that the left C^∞ module of derivation of the ring $C^\infty(M)$ i.e. linear maps $D : C^\infty(M) \to C^\infty(M)$ satisfying

$$D(fg) = (Df)g + f(Dg)$$

can be identified with vector fields on M, smooth sections of TM.

Proof. Define

$$D(x)f = (Df)(x)$$

Then

$$D(x)(fg) =$$

□

1 Sard’s Theorem